

String Theory and M-Theory

Lecture 4: Closed Strings, Level Matching, and Duality

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Chapter 1

Closed Strings, Level Matching, and Duality

Overview. A closed string supports independent waves in its two intrinsic directions. Quantizing those waves seems at first to double the open-string photon, but invariance under shifting the origin of the string coordinate removes the one-sided states. The first surviving level contains a massless spin-two particle. The same idea of changing descriptions then opens a broader question: when couplings vary, what does it mean for any object to be fundamental?

1.1 A Generator Kept in Reserve

Before drawing a closed string, let us recover one fact from canonical mechanics. Suppose

$$L = L(q_i, \dot{q}_i), \quad (1.1)$$

and define the momentum conjugate to q_i by

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \quad (1.2)$$

An infinitesimal continuous transformation can be written

$$\delta q_i = \epsilon f_i(q), \quad (1.3)$$

where ϵ is a small parameter. If this transformation is a symmetry, Noether's theorem supplies a conserved charge. Dividing the coordinate displacement by the common parameter gives the generator

$$Q = \sum_i p_i f_i(q) = \frac{1}{\epsilon} \sum_i p_i \delta q_i. \quad (1.4)$$

In quantum mechanics, Q is the operator that generates the corresponding small transformation.

A rotation in the xy -plane is the familiar example. With one choice of orientation,

$$\delta x = \epsilon y, \quad \delta y = -\epsilon x, \quad (1.5)$$

and (1.4) becomes angular momentum, up to the same choice of sign. The particular example will not matter. What matters is the pattern: momentum contracted with the infinitesimal displacement gives the generator.

The image shows two handwritten equations on a light-colored background. The top equation is $P_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$. The bottom equation is $Q = \sum_i P_i f_i(\tau)$, with a circle around the Q and the word "Noether" written below it.

Figure 1.1: The canonical momentum and Noether generator retained for the later shift of the closed-string coordinate.

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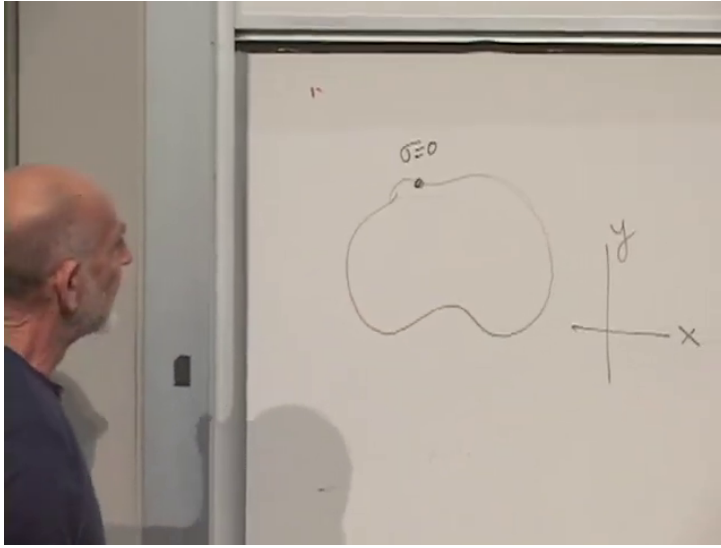
We shall shortly encounter a symmetry whose coordinate is not a point in ordinary space but the labeling of points along a string. Equation (1.4) is being placed within reach for that purpose.

1.2 A Coordinate Around a Loop

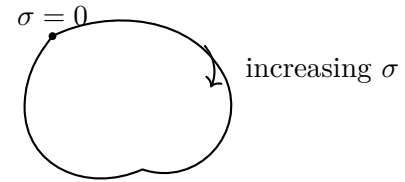
A closed string has no endpoints. Choose a parameter

$$0 \leq \sigma < 2\pi \tag{1.6}$$

and, for the moment, mark one point as $\sigma = 0$. Halfway around is $\sigma = \pi$; a quarter-turn and three-quarter-turn are $\pi/2$ and $3\pi/2$. The mark establishes a coordinate convention, not yet a physical feature.



(a) The marked loop on the blackboard.



(b) The intrinsic orientation.

Figure 1.2: A point is marked to define the origin, and an arrow fixes the direction in which σ increases.

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The arrow along the loop is an *intrinsic* orientation. It must not be confused with clockwise or counterclockwise motion in the xy -plane. Imagine a marked rubber band lifted from the board, turned over, and put back. Its apparent clockwise sense reverses, while the sequence of its intrinsic marks does not. Increasing σ remains increasing σ .

This distinction controls the language of right- and left-moving waves. In this chapter, a right-moving disturbance means one traveling toward increasing σ , and a left-moving disturbance means one traveling toward decreasing σ . Neither name tells us which way the disturbance appears to move in the room.

Orientation had a concrete role in the original hadronic string picture. Breaking an oriented flux tube produces a quark at one endpoint and an antiquark at the other; the intrinsic arrow distinguishes the two ends. One can also construct unoriented string theories, but for the present discussion we retain the arrow and the two senses along it.

1.3 Periodicity and Running Waves

Let $X(\sigma, \tau)$ and $Y(\sigma, \tau)$ be two transverse coordinates of the string, with τ denoting time. Extra transverse coordinates, when present, are treated in exactly the same way. Since $\sigma = 0$ and $\sigma = 2\pi$ label the same point,

$$X(\sigma + 2\pi, \tau) = X(\sigma, \tau), \quad Y(\sigma + 2\pi, \tau) = Y(\sigma, \tau). \quad (1.7)$$

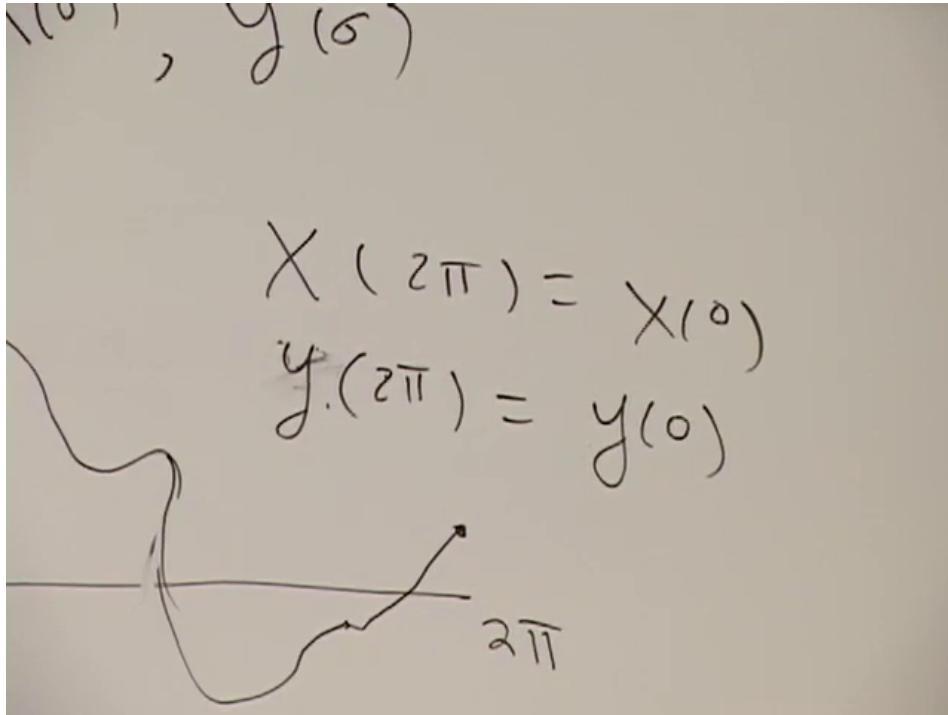


Figure 1.3: Periodic boundary conditions for the two transverse coordinates.

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A localized wiggle can therefore circulate indefinitely. When it reaches 2π , it simply reappears at 0. This differs from an open string, where a running wave reaches an endpoint and reflects; what was left-moving becomes right-moving and vice versa. Standing waves are natural for the open string, whereas a closed string admits two independently propagating families.

1.3.1 Complex Fourier modes

The convenient basis for a periodic function is

$$e^{in\sigma} = \cos(n\sigma) + i\sin(n\sigma), \quad n \in \mathbb{Z}. \quad (1.8)$$

The integer condition is forced by periodicity: $e^{in(\sigma+2\pi)} = e^{in\sigma}$. Replacing n by a half-integer, for example, would change the function after one circuit.

Suppressing time dependence momentarily, the blackboard expansion is

$$X(\sigma) = X_0 + \sum_{n>0} X_n e^{in\sigma} + \sum_{n>0} X_{-n} e^{-in\sigma}, \quad (1.9)$$

$$Y(\sigma) = Y_0 + \sum_{n>0} Y_n e^{in\sigma} + \sum_{n>0} Y_{-n} e^{-in\sigma}. \quad (1.10)$$

For real X and Y , the negative-index coefficients are complex conjugates of the positive-index coefficients at the classical level. The zero modes X_0 and Y_0 locate the center of mass; they are positions, not internal oscillations.

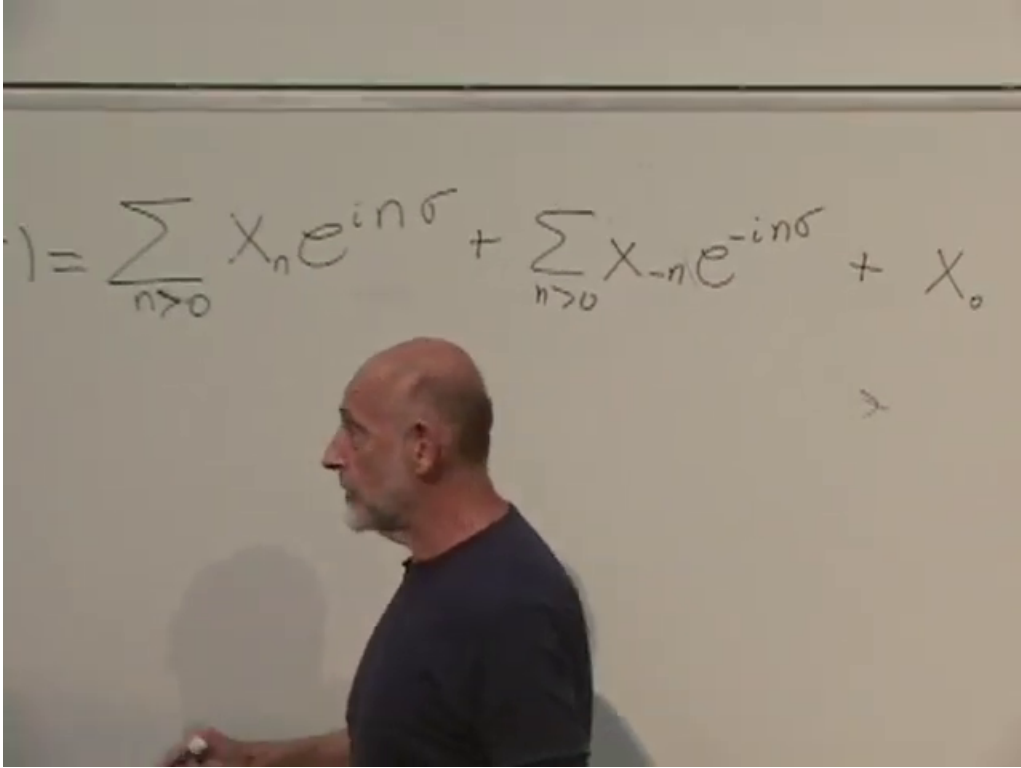


Figure 1.4: The positive- and negative-index pieces of the periodic Fourier expansion, followed by the zero mode.

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Question from the audience. If these are moving waves, where is the time coordinate?^a

Response. It has only been suppressed in the board notation. A traveling solution carries a phase involving both σ and τ . With the convention used below, one family has $e^{-in(\tau-\sigma)}$ and the other $e^{-in(\tau+\sigma)}$.

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Thus it is the full phase, not the sign of n in a purely spatial Fourier factor by itself, that determines propagation. We may write

$$X(\sigma, \tau) = X_R(\tau - \sigma) + X_L(\tau + \sigma), \quad (1.11)$$

and similarly for Y . Which family is called R rather than L is a convention; the invariant fact is that there are two opposite directions along the σ -circle.¹

1.4 Energy in the Two Directions

The local dynamics is the same as for the open string. In the simplified normalization used here,

$$L \propto \int_0^{2\pi} d\sigma \left[\dot{X}^2 - X'^2 + \dot{Y}^2 - Y'^2 \right], \quad (1.12)$$

¹Editorial clarification: A factor $e^{in\sigma}$ at one instant does not by itself specify a moving wave. Propagation is fixed only after the time phase is supplied.

where a dot denotes ∂_τ and a prime denotes ∂_σ . The energy has a plus sign between kinetic and elastic terms:

$$E \propto \int_0^{2\pi} d\sigma \left[\dot{X}^2 + X'^2 + \dot{Y}^2 + Y'^2 \right]. \quad (1.13)$$

For either transverse coordinate,

$$\dot{X}^2 + X'^2 = \frac{1}{2}(\dot{X} + X')^2 + \frac{1}{2}(\dot{X} - X')^2. \quad (1.14)$$

The cross terms cancel, leaving precisely the original kinetic and gradient energies.

$$E = \int_0^{2\pi} d\sigma \left(\left(\frac{\partial X}{\partial \tau} \right)^2 + \left(\frac{\partial X}{\partial \sigma} \right)^2 \right)$$

$$= \frac{1}{2} \left(\frac{\partial X}{\partial \tau} + \frac{\partial X}{\partial \sigma} \right)^2 + \frac{1}{2} \left(\frac{\partial X}{\partial \tau} - \frac{\partial X}{\partial \sigma} \right)^2$$

Figure 1.5: The total energy reorganized into the two running-wave sectors.

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For $X_R(\tau - \sigma)$, one has $\dot{X}_R = -X'_R$; for $X_L(\tau + \sigma)$, one has $\dot{X}_L = X'_L$. Consequently the two squares in (1.14) isolate the two chiral families. Including every transverse coordinate, define

$$E_L \propto \frac{1}{2} \int_0^{2\pi} d\sigma \sum_I (\dot{X}^I + X'^I)^2, \quad (1.15)$$

$$E_R \propto \frac{1}{2} \int_0^{2\pi} d\sigma \sum_I (\dot{X}^I - X'^I)^2. \quad (1.16)$$

Then $E = E_L + E_R$. Because the equations are linear at this stage, waves from the two sectors pass through one another without scattering.

1.4.1 Twice as many oscillators

Substituting the Fourier series into (1.12) turns each nonzero mode coefficient into a harmonic oscillator. Its frequency is proportional to $|n|$: shorter wavelength means higher frequency and therefore a larger excitation energy. Unlike the open string, each positive frequency occurs in two independently propagating copies.

The blackboard labels the X -polarized creation operators a_n^+ and a_{-n}^+ , and the Y -polarized ones b_n^+ and b_{-n}^+ . To separate creation from propagation more cleanly, we shall write

$$a_n^+, \quad b_n^+, \quad \tilde{a}_n^+, \quad \tilde{b}_n^+, \quad n > 0. \quad (1.17)$$

The untilded and tilded operators create opposite world-sheet directions; a and b denote the two transverse polarizations.

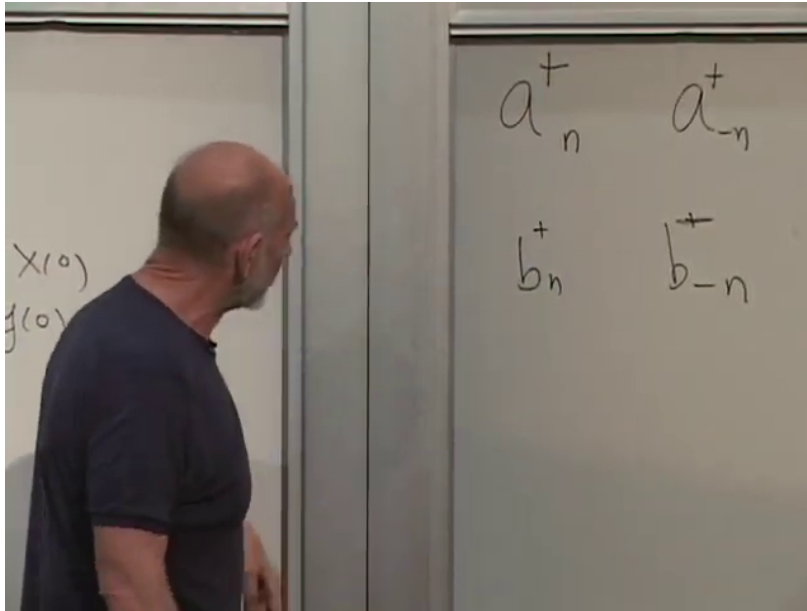


Figure 1.6: The board notation for the doubled X - and Y -polarized creation operators.

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If $N_n^{(a)}, N_n^{(b)}$ and their tilded partners are occupation numbers, the oscillator levels carried in the two directions are schematically

$$\begin{aligned} N &= \sum_{n>0} n(N_n^{(a)} + N_n^{(b)} + \cdots), \\ \tilde{N} &= \sum_{n>0} n(\tilde{N}_n^{(a)} + \tilde{N}_n^{(b)} + \cdots). \end{aligned} \quad (1.18)$$

The ellipses stand for additional transverse directions.

1.5 The Spectrum Guess That Fails

Recall the open-string first level. Acting on the ground state with $a_1^\dagger$ or $b_1^\dagger$ gives two linear polarizations. Their circular combinations,

$$c_\pm^\dagger = \frac{1}{\sqrt{2}} (a_1^\dagger \pm ib_1^\dagger), \quad (1.19)$$

carry helicity $h = \pm 1$. The absence of a helicity-zero partner is the signature of a massless vector in this elementary four-dimensional description.

Let the closed-string ground state have mass squared m_0^2 . A first, naive guess is to act once with any one of

$$a_1^\dagger, \quad b_1^\dagger, \quad \tilde{a}_1^\dagger, \quad \tilde{b}_1^\dagger. \quad (1.20)$$

This resembles two copies of the open-string photon, one for each propagation direction. At the next energy, however, the unrestricted list does not assemble into sensible rotational multiplets. The problem is not repaired by more elaborate bookkeeping. The premise that every oscillator monomial is a physical state is wrong.

1.5.1 Level matching, first as a rule

The physical closed-string condition is

$$N = \tilde{N}, \quad \text{equivalently} \quad E_L = E_R. \quad (1.21)$$

For the moment, take this as a postulate; its origin will be derived only after we have seen what it accomplishes.

Every state in (1.20) has one unit of energy in one sector and none in the other, so every one is excluded. The first surviving states carry one oscillator from each family:

$$a_1^\dagger \tilde{a}_1^\dagger |0\rangle, \quad b_1^\dagger \tilde{b}_1^\dagger |0\rangle, \quad (1.22)$$

$$a_1^\dagger \tilde{b}_1^\dagger |0\rangle, \quad b_1^\dagger \tilde{a}_1^\dagger |0\rangle. \quad (1.23)$$

They have total oscillator energy two, divided equally between the two directions. A single $n = 2$ creation operator has the same total oscillator energy but fails (1.21); it contributes two units to only one side.

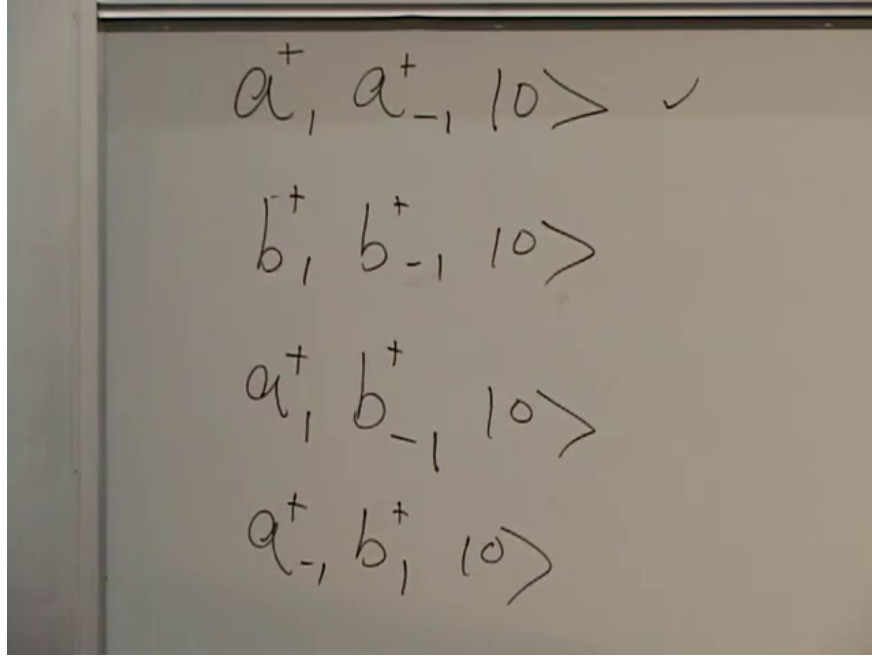


Figure 1.7: The four first-level states that survive level matching. The board uses subscripts 1 and -1 where (1.22)–(1.23) use untilded and tilded operators.

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1.6 A Graviton in the First Matched Level

Put both chiral families into a circular basis:

$$c_{\pm}^{\dagger} = \frac{a_1^{\dagger} \pm i b_1^{\dagger}}{\sqrt{2}}, \quad \tilde{c}_{\pm}^{\dagger} = \frac{\tilde{a}_1^{\dagger} \pm i \tilde{b}_1^{\dagger}}{\sqrt{2}}. \quad (1.24)$$

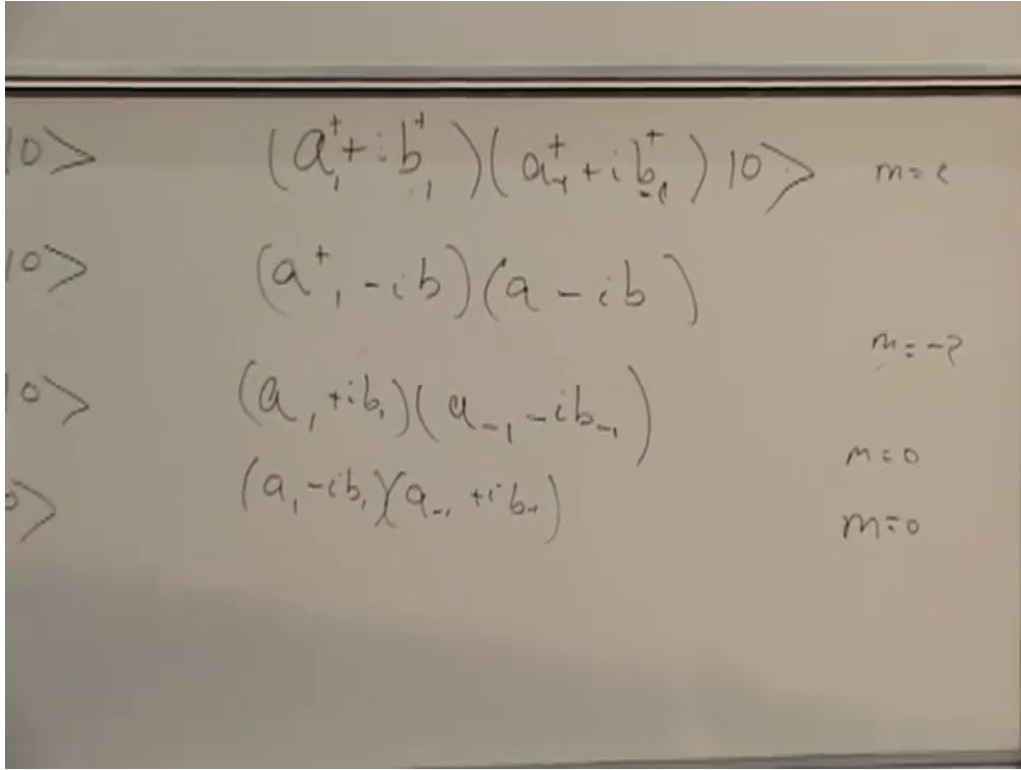
Each factor contributes helicity $+1$ or -1 in ordinary transverse space. Their products therefore give

State	Helicity
$c_+^{\dagger} \tilde{c}_+^{\dagger} 0\rangle$	$+2$
$c_-^{\dagger} \tilde{c}_-^{\dagger} 0\rangle$	-2
$c_+^{\dagger} \tilde{c}_-^{\dagger} 0\rangle$	0
$c_-^{\dagger} \tilde{c}_+^{\dagger} 0\rangle$	0

Question from the audience. Should the second circular-polarization factor come from the opposite propagation sector?^a

Response. Yes. The first expression written on the board was corrected for precisely this reason. A physical state here must pair one right-moving oscillator with one left-moving oscillator; otherwise it is not level matched. The helicities refer to transverse-space polarization, while the two factors refer to opposite world-sheet directions.

^aLecture timestamp: 00:43:20.

Figure 1.8: The four circular-basis products after the board correction, with helicities $+2, -2, 0, 0$.*Lecture frame: 00:46:10.*

A massive spin-two particle has five spin projections,

$$m = 2, 1, 0, -1, -2. \quad (1.25)$$

The matched level contains no $m = \pm 1$ partners. It therefore cannot be a massive spin-two multiplet. A massless spin-two particle, by contrast, has only helicities $+2$ and -2 . The same-helicity products are exactly those two states: they are the graviton polarizations.

Two helicity-zero combinations remain. In the language used in class, one is the dilaton and the other the axion. A covariant description of the generic closed bosonic string refines this statement:

$$\alpha_{-1}^I \tilde{\alpha}_{-1}^J |0; k\rangle \longrightarrow \underbrace{h_{IJ}}_{\text{symmetric traceless}} \oplus \underbrace{B_{IJ}}_{\text{antisymmetric}} \oplus \underbrace{\Phi}_{\text{trace}}. \quad (1.26)$$

These are the graviton, antisymmetric two-form, and dilaton. In four dimensions the two-form can be dualized to an axion-like pseudoscalar, which is the language adopted in the lecture.²

At this formal level the three fields are massless. Compactification, projections, and background dynamics can alter which scalar or two-form degrees of freedom remain light. The graviton is different: once a consistent interacting closed-string sector is present, its massless spin-two state is unavoidable. This is why gravity emerges from closed strings rather than being appended as a separate ingredient.

²Editorial clarification: The second helicity-zero state is fundamentally the antisymmetric tensor field $B_{\mu\nu}$ in the generic closed-string spectrum. Calling it an axion is the four-dimensional dual description.

1.7 Why the Levels Must Match

We now return to the mark at $\sigma = 0$. Is it a physical bead attached to the loop, or merely the origin of our parameter? For an ordinary unmarked closed string, no point should be privileged. The quantum state must be unchanged when every label is shifted together.

1.7.1 The discrete version

Regulate the loop as M ordered points with transverse coordinates (X_i, Y_i) . Temporarily suppress Y_i . A configuration-space wavefunction is

$$\Psi(X_1, X_2, \dots, X_M). \quad (1.27)$$

Changing which point is called first gives a cyclic permutation:

$$\begin{aligned} \Psi(X_1, X_2, \dots, X_M) &= \Psi(X_2, \dots, X_M, X_1) \\ &= \Psi(X_3, \dots, X_M, X_1, X_2). \end{aligned} \quad (1.28)$$

This is not invariance under an arbitrary permutation. Inserting X_3 between X_1 and X_2 , for example, changes the adjacency structure and effectively tears and reconnects the discretized string. Only the common cyclic relabeling

$$X_i \longrightarrow X_{i+1} \quad (\text{mod } M) \quad (1.29)$$

expresses the absence of a preferred origin.

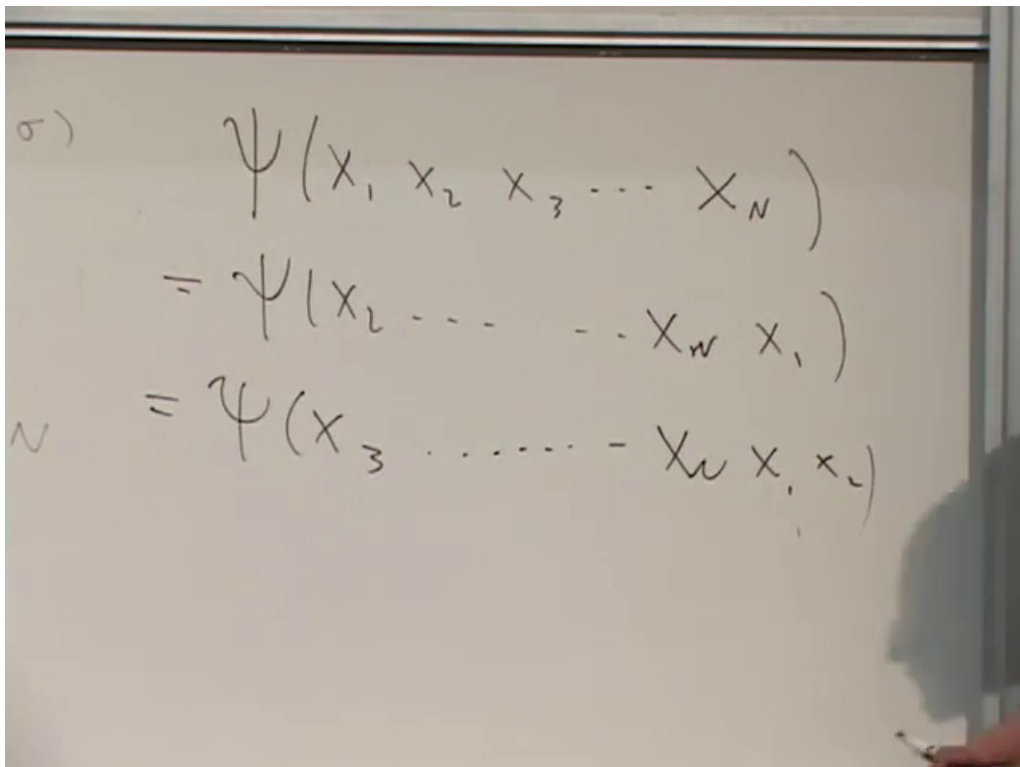


Figure 1.9: The cyclic wavefunction identities for a discretized closed string.

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1.7.2 The continuum version

In the continuum limit $M \rightarrow \infty$, the state is the wavefunctional

$$\Psi[X(\sigma), Y(\sigma)].$$

A cyclic relabeling becomes

$$\sigma \longrightarrow \sigma + \epsilon. \quad (1.30)$$

Invariance says

$$\Psi[X(\sigma + \epsilon), Y(\sigma + \epsilon)] = \Psi[X(\sigma), Y(\sigma)]. \quad (1.31)$$

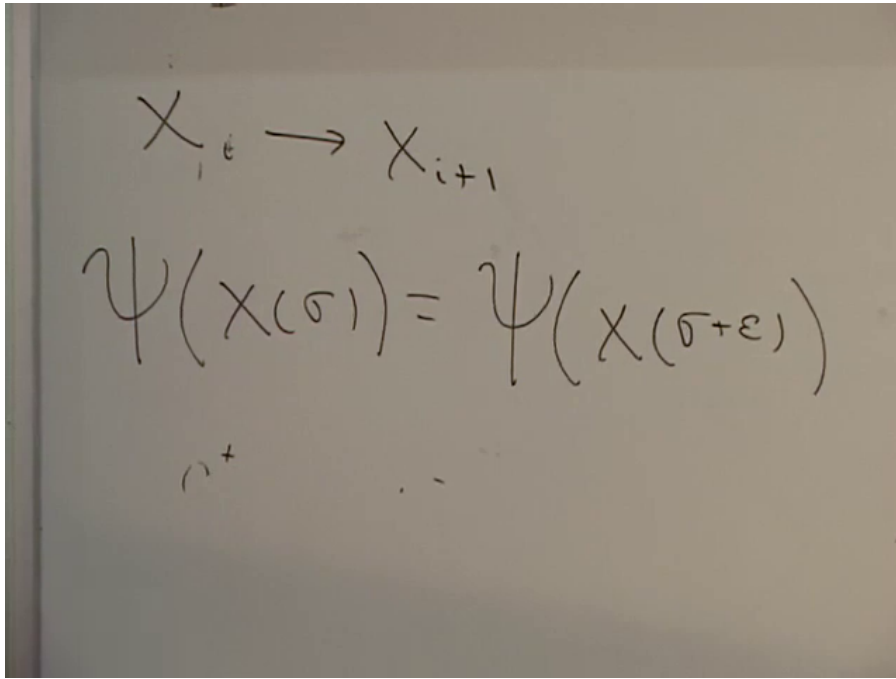


Figure 1.10: The discrete shift and its wavefunctional continuum form.

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To first order,

$$\delta X(\sigma) = \epsilon X'(\sigma), \quad \delta Y(\sigma) = \epsilon Y'(\sigma), \quad (1.32)$$

up to the overall sign that distinguishes an active shift of the profile from a passive shift of the coordinate. The corresponding wavefunctional variation is

$$\delta \Psi = \epsilon \int_0^{2\pi} d\sigma \left[\frac{\delta \Psi}{\delta X(\sigma)} X'(\sigma) + \frac{\delta \Psi}{\delta Y(\sigma)} Y'(\sigma) \right]. \quad (1.33)$$

$$\begin{aligned} \circ &= \psi(X(\sigma + \epsilon)) - \\ &\quad - \frac{\partial \psi}{\partial X(\sigma)} \frac{\partial X}{\partial \sigma} \epsilon \end{aligned}$$

Figure 1.11: The first-order expansion of the shifted wavefunctional argument.

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Question from the audience. Is the first-order condition a separate equation for each value of σ ?^a

Response. No. The transformation shifts every point of the string together. The variation at one point must be added to the variations at all other points, which is why the continuum expression contains the integral over σ .

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In the coordinate representation,

$$P_X(\sigma) = -i \frac{\delta}{\delta X(\sigma)}, \quad P_Y(\sigma) = -i \frac{\delta}{\delta Y(\sigma)} \quad (1.34)$$

with $\hbar = 1$. Equation (1.33) is therefore generated by

$$G_\sigma = \int_0^{2\pi} d\sigma [P_X(\sigma) X'(\sigma) + P_Y(\sigma) Y'(\sigma)]. \quad (1.35)$$

This is exactly the Noether pattern (1.4): canonical momentum multiplied by the coordinate displacement, summed over all degrees of freedom. An unmarked closed-string state obeys

$$G_\sigma |\text{phys}\rangle = 0. \quad (1.36)$$

For the normalization of (1.12), $P_I \propto \dot{X}^I$, so the same condition reads

$$\int_0^{2\pi} d\sigma \sum_I \dot{X}^I X^{I'} = 0. \quad (1.37)$$

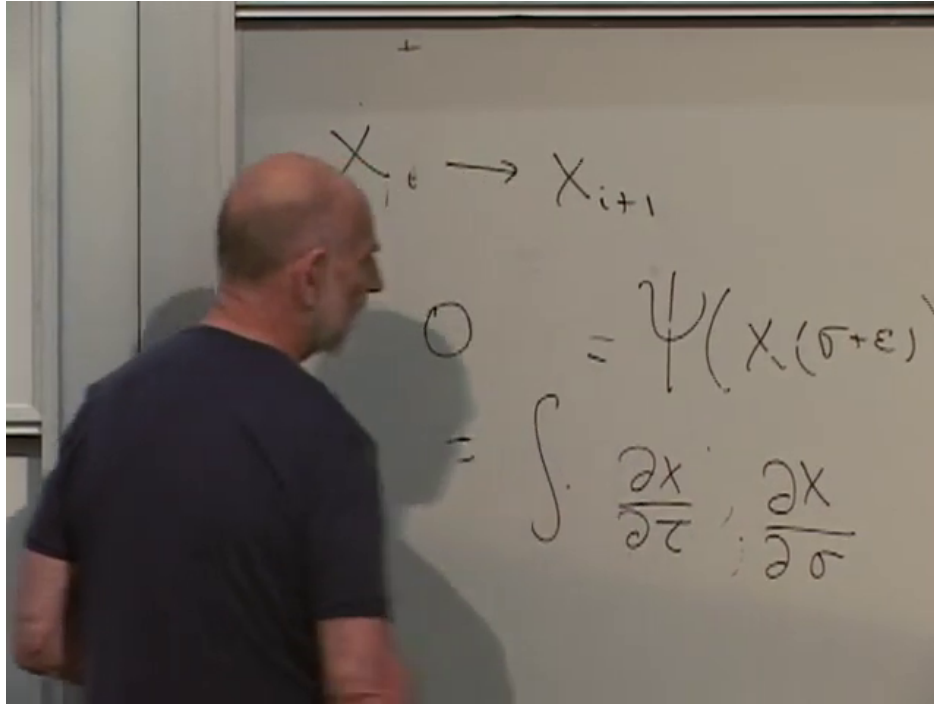


Figure 1.12: The integrated generator after the derivative of the wavefunctional is identified with momentum.

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1.7.3 The generator is the energy difference

The sum of the two expressions in (1.16) is the total energy. Their difference is

$$E_L - E_R \propto \frac{1}{2} \int_0^{2\pi} d\sigma \sum_I \left[(\dot{X}^I + X'^I)^2 - (\dot{X}^I - X'^I)^2 \right] \quad (1.38)$$

$$= 2 \int_0^{2\pi} d\sigma \sum_I \dot{X}^I X'^I. \quad (1.39)$$

The integral is the generator that must annihilate a physical state. Therefore

$$E_L - E_R = 0, \quad \text{or} \quad N - \tilde{N} = 0. \quad (1.40)$$

Level matching is the statement that the origin of the σ -coordinate is unobservable.

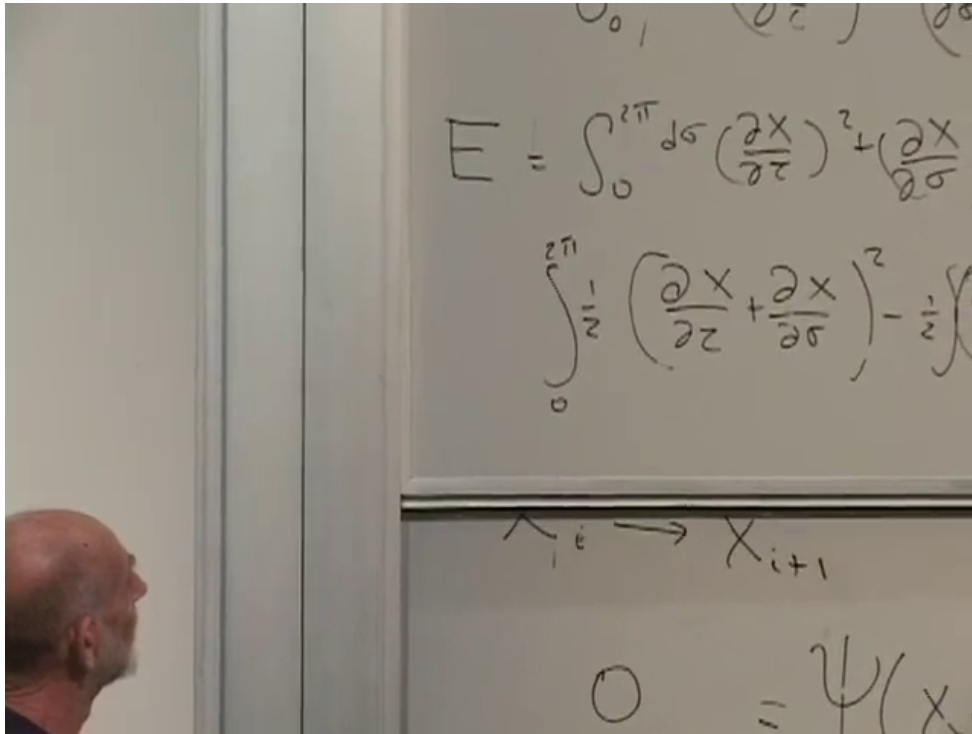


Figure 1.13: The energy sum and difference displayed together at the end of the level-matching derivation.

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Question from the audience. Why do the displayed formulas appear to place the factor of one-half differently?^a

Response. Only the relative structure is being used. Restoring one consistent overall normalization gives the one-half in the completed-square identity (1.14). The sum yields total energy, while the difference remains proportional to $\int d\sigma \dot{X} X'$. Its vanishing, and hence level matching, is independent of that common convention.

^aLecture timestamp: 01:06:28.

The constraint removes a large portion of the naive oscillator spectrum, and it removes precisely the states that failed to form sensible particle multiplets. It does not remove the open-string photon, because an open string does not possess two independent circulating sectors subject to this closed loop condition. For the closed string, it leaves the graviton, the dilaton, the antisymmetric-tensor or axion-like state, and the higher matched levels.³

1.8 Two Meanings of Left and Right

There are now two unrelated uses of handed language:

³Editorial clarification: In covariant string quantization, level matching is the zero-mode condition $L_0 - \tilde{L}_0 = 0$, one member of the world-sheet constraint system. The derivation here isolates its intuitive origin as invariance under translations around the closed string.

1. $X_R(\tau - \sigma)$ and $X_L(\tau + \sigma)$ describe propagation toward opposite directions along the intrinsic σ -circle.
2. $c_+^\dagger$ and $c_-^\dagger$ describe circular polarization in the physical transverse xy -plane.

The level-matched graviton state contains one oscillator from each world-sheet direction. Those two oscillators may nevertheless have the same spacetime helicity, producing $h = +2$ or $h = -2$. Confusing these two kinds of left and right makes the spectrum look contradictory when it is not.

1.9 What Does Fundamental Mean?

Once open and closed strings have been quantized at this elementary level, a natural question remains: are strings the final constituents, or could they themselves be made of something else?

Question from the audience. Is there evidence or a reason to regard strings as the most fundamental objects, with nothing smaller underneath them?^a

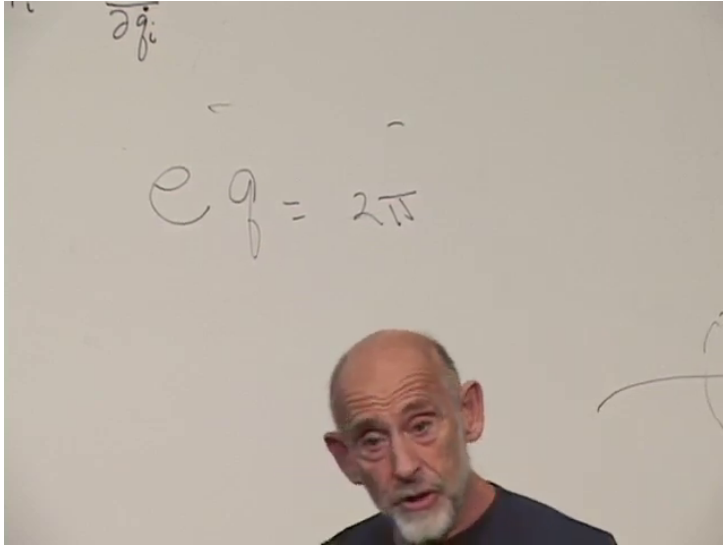
Response. There is no useful parameter-independent answer. The degrees of freedom that look elementary in one regime can become heavy, extended, and apparently composite in another. Fundamental is best read as a statement about which variables provide the useful description at the coupling under consideration.

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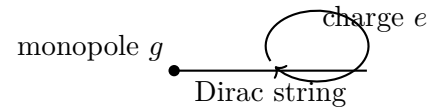
Electric and magnetic charges provide the cleanest model of this answer. Suppose a theory contains an electric charge e and a magnetic charge g . The unobservable Dirac string attached to a monopole requires

$$eg = 2\pi n, \quad n \in \mathbb{Z}. \quad (1.41)$$

For the minimal charges, $n = 1$. A charged particle circling the Dirac string then acquires a phase $e^{ieg} = 1$, so the auxiliary string cannot be detected.



(a) Dirac quantization on the board.



(b) The phase experiment around the string.

Figure 1.14: Electric and magnetic charges are inversely related by the Dirac condition.

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If e is small, ordinary perturbation theory in electric variables is useful. Feynman diagrams with additional electric vertices are increasingly suppressed. But (1.41) then makes g large, so a diagrammatic expansion in magnetic vertices is not useful. In terms of the usual dimensionless couplings,

$$\alpha_e = \frac{e^2}{4\pi}, \quad \alpha_m = \frac{g^2}{4\pi}, \quad \alpha_e \alpha_m = \frac{n^2}{4}. \quad (1.42)$$

Weak electric coupling is strong magnetic coupling, and conversely.

The strongly coupled monopole also looks complicated from the electric description. Its field energy is large; it is dressed by an intense cloud of gauge-field fluctuations and virtual magnetically charged pairs. It behaves as a heavy, extended object, a poor starting point for an electric perturbation series. This does not prove that it is less fundamental. It says that it is the wrong elementary variable for that regime.

Now imagine continuously increasing e . The quantization condition forces g to decrease. Near the self-dual region neither description is especially weak, and beyond it the magnetic variables can become the simpler, lighter ones while the electric objects become strongly dressed. Which charge should be called elementary has changed without changing the theory's physical content.

1.9.1 The Wigner question

This ambiguity once surfaced in a discussion of whether the Higgs boson was fundamental or composite. Eugene Wigner repeatedly asked what composite meant. One proposed distinction was decay: perhaps a fundamental object cannot decay and a composite object can. But a ground-state hydrogen atom cannot decay and is plainly composite; the proton may be stable while the neutron decays, yet that does not make one elementary and the other composite in the intended sense. Every simple definition generated another counterexample.

Gerardus 't Hooft ended the discussion with the operational criterion: an object is fundamental when it is useful to think of it as fundamental. This is not an evasion. It recognizes that elementary identifies the variables in which the dynamics is simple, and simplicity can depend on coupling and environment.

1.10 Toward M-Theory

String theory realizes this exchange of roles in a particularly sharp way. In one parameter regime, strings are light and perturbative, while D-branes look like heavy solitonic composites built from string degrees of freedom. As the parameters change, D-branes can become the lighter elementary objects, and configurations described as strings in the first regime can be reinterpreted in terms of them. What looked fundamental can become composite; what looked composite can become fundamental.

The old reductionist picture suggests a one-way descent from large things to smaller constituents and finally to a unique bottom layer. Duality suggests something less linear: descriptions can transform into one another as couplings vary, with no single set of variables privileged everywhere. That is the angle from which the next subject, M-theory, will be approached.

The questions of why the critical dimensions are twenty-six for the bosonic string and ten for the superstring remain open here. Rather than settling every technical issue before moving on, we now change viewpoint and ask how M-theory can produce string theory in an appropriate limit.